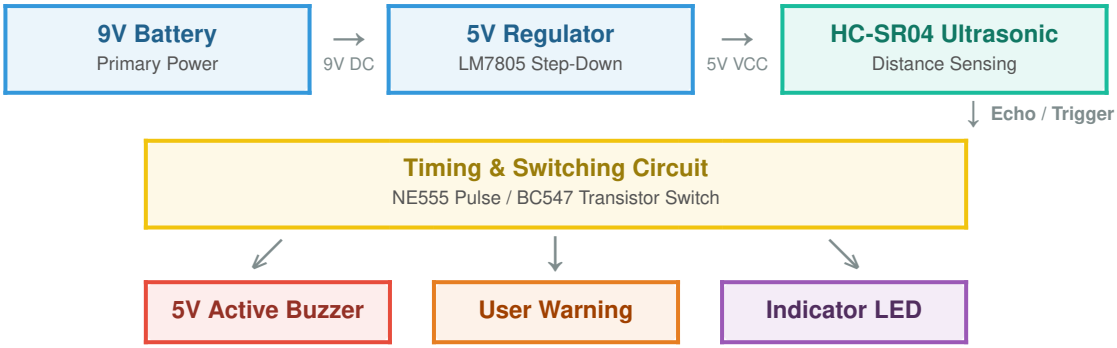
SMART BLIND STICK

An Affordable, Microcontroller-Free Assistive Mobility Device

1. METHODOLOGY

1. **Mounting:** An HC-SR04 ultrasonic sensor is securely fixed to the upper front portion of the walking stick, securing a clear, wide field of view for uninterrupted obstacle scanning.
2. **Regulated Power:** A standard 9V battery powers the unit, stepped down via a 5V voltage regulator (LM7805) to ensure a steady, safe energy supply for all sensitive electronic parts.
3. **Signal Triggering:** A discrete timing circuit generates periodic 10 µs trigger pulses, prompting the sensor to reliably emit 40 kHz ultrasonic sound bursts.
4. **Echo Detection & Calculation:** Emitted pulses bounce off physical barriers and return. The sensor's echo pin outputs a high pulse directly proportional to the round-trip travel time.
5. **Switching & Warning:** A BC547 transistor operates as a high-speed switch. When an obstacle enters the hazard threshold, it instantly triggers a 5V active buzzer and a high-intensity alert LED, warning the user.

2. PROPOSED BLOCK DIAGRAM



3. KEY ADVANTAGES

- **Ultra Low-Cost & Accessible:** By eliminating expensive microcontrollers and complex processors, the manufacturing cost is kept incredibly low. This ensures the device is highly affordable so that everyone in the public can buy it easily without financial strain.
- **Designed for the General Public:** It serves as a universal, highly accessible mobility aid for the visually impaired. It is user-friendly and requires absolutely zero technical setup, programming, or specialized training to operate safely.
- **High Reliability & Instant Feedback:** Relying strictly on discrete hardware switching (transistors and timers) means there is no software to crash, lag, or freeze, guaranteeing immediate, fail-proof audible and visual alerts.
- **Easily Retrofittable & Lightweight:** The compact circuit and sensor module add minimal weight and can be easily retrofitted onto any standard walking cane, upgrading existing equipment instantly.

4. PROPOSED OUTCOME

The proposed Smart Blind Stick delivers an affordable, dependable, and highly portable walking aid engineered to promote safe and independent mobility. It successfully achieves instantaneous, real-time obstacle detection up to 400 cm with rapid audible and visual warnings, offering a highly accessible, low-maintenance, and impactful assistive technology solution for the public.